

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-004: Student Input Field Validation for Add and Edit Operations

Student Management System

Field	Details
Change Request ID	CR-004
Change Request Title	Student Input Field Validation for Add and Edit Operations
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	Completed
Date	8 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators previously could enter student information without sufficient validation during Add and Edit operations. The change was required to improve student data quality and prevent invalid or insecure information from being submitted or stored. Validation was required for registration number, first name, last name, email address, phone number, and new-student password.

1.2 Existing Behaviour

Before implementation:

- StudentFormModal.jsx provided only basic required-field checking.
- Phone numbers were not comprehensively restricted to exactly 10 numeric digits.
- First and last names did not enforce a minimum of 3 characters or valid character rules.
- Email validation was insufficient.
- Registration number validation was not comprehensive.
- New student passwords did not enforce the required strong-password policy.
- Backend validation did not comprehensively validate all student input.
- Invalid data could potentially be submitted directly to the backend.

1.3 Requested Change

The following validation was required for Student Add and Edit operations:

- Validate all required fields before submission.
- Validate the registration number.
- Require first and last names to contain at least 3 characters.
- Allow valid alphabetic name characters and reject numbers/inappropriate special characters.

- Validate email address format.
- Require the phone number to contain exactly 10 numeric digits.
- Require a new-student password to contain at least 8 characters, uppercase, lowercase, number, and special character.
- Display clear validation errors.
- Prevent invalid frontend submission.
- Apply server-side validation to prevent direct API bypass.
- Preserve existing Add, Edit, Delete, authentication, authorization, and password hashing behaviour.

2. Implementation

2.1 Implementation Summary

The CR-004 requirements were implemented in the Student Add/Edit functionality. Client-side validation was added to the student form to provide immediate feedback and prevent invalid submission. Server-side validation was added to the existing student controller so that invalid requests are also rejected when submitted directly to the API. The existing CRUD flow, authentication/authorization behaviour, and password hashing mechanism were preserved.

2.2 Affected Files

File	Change	Purpose
frontend/src/components/StudentFormModal.jsx	Modified	Implemented client-side validation, validation messages, and Add/Edit submission blocking.
backend/controllers/studentController.js	Modified	Implemented server-side validation for Student Add/Edit requests and prevented invalid API input.

2.3 Technical Changes

Required Field Validation

- Required student fields are checked before submission.
- Missing required values are rejected with clear validation feedback.

Registration Number Validation

- Registration number is validated before Add/Edit submission.
- Invalid or missing registration-number input is rejected.

Name Validation

- First and last names must contain at least 3 characters.
- Invalid numeric or inappropriate special characters are rejected.
- The same rules are applied during Add and Edit operations.

Email Validation

- Email input is checked against a valid email format.
- Invalid email values are rejected before submission.

Phone Validation

- Phone number must contain exactly 10 digits.
- Only numeric characters are accepted.
- Shorter, longer, or non-numeric phone values are rejected.

Strong Password Validation

- A password is required when creating a new student.
- Password must contain at least 8 characters.
- Password must include an uppercase letter.
- Password must include a lowercase letter.
- Password must include a number.
- Password must include a special character.
- The existing password hashing mechanism is preserved.

Client-Side and Server-Side Validation

- Frontend validation provides immediate administrator feedback.
- Backend validation prevents invalid requests from bypassing the frontend.
- Invalid API input is rejected before persistence.

2.4 Implementation Logic

1. Administrator opens the Add Student or Edit Student form.
2. Frontend checks required fields and validation rules.
3. If invalid input is detected, validation errors are displayed and submission is stopped.
4. Valid input is sent through the existing Student Add/Edit API flow.
5. Backend validates the received request again.
6. If backend validation fails, the request is rejected and invalid data is not stored.
7. If validation succeeds, the existing CRUD operation continues.
8. Existing authentication, authorization, and password hashing behaviour remains unchanged.

2.5 Error and Data Handling

The implementation provides clear validation feedback for invalid student input. Frontend validation prevents incorrect data from being submitted through the form, while backend validation provides the final validation boundary for direct API requests. Invalid values are rejected before persistence. Existing error handling outside the validation scope remains unchanged.

3. Testing & Verification

3.1 Testing Approach

The implementation was verified through manual frontend and backend validation testing covering positive, negative, boundary, API, security, and regression scenarios.

Testing Types:

- Functional testing
- Positive testing
- Negative testing
- Boundary testing
- Validation testing
- UI testing
- Backend/API testing
- Security testing
- Regression testing
- CRUD verification

Environment: Development / Test

Browser: Chrome / Web Browser

Test Role: Administrator

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Submit with empty required fields	Required-field errors are displayed and submission is blocked	Validation blocked	Pass
TC-02	First name with fewer than 3 characters	First name is rejected	Input rejected	Pass
TC-03	Last name with fewer than 3 characters	Last name is rejected	Input rejected	Pass
TC-04	First/last name containing invalid characters	Invalid name is rejected	Input rejected	Pass
TC-05	Invalid email address	Email is rejected	Input rejected	Pass
TC-06	Phone number shorter than 10 digits	Phone number is rejected	Input rejected	Pass
TC-07	Phone number longer than 10 digits	Phone number is rejected	Input rejected	Pass
TC-08	Phone number containing non-numeric characters	Phone number is rejected	Input rejected	Pass
TC-09	Valid 10-digit phone number	Phone number is accepted	Valid phone accepted	Pass

TC-10	Weak password	Password is rejected	Weak password rejected	Pass
TC-11	Strong password	Password is accepted	Strong password accepted	Pass
TC-12	Valid student creation	Student is created successfully	Student created	Pass
TC-13	Invalid student creation	Invalid data is blocked	Creation blocked	Pass
TC-14	Invalid student update	Invalid edited data is blocked	Update blocked	Pass
TC-15	Valid student update	Student is updated successfully	Student updated	Pass
TC-16	Direct API request with invalid data	Backend rejects invalid request	Request rejected	Pass
TC-17	Existing Edit/Delete operations	Correct student remains affected by student_id	CRUD worked correctly	Pass
TC-18	Existing administrator authentication/authorization	Existing access control continues to work	Access control preserved	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	18
Passed	18
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Empty required fields are rejected.	Required-field validation test	Met
AC-002	First name with fewer than 3 characters is rejected.	Name boundary test	Met
AC-003	Last name with fewer than 3 characters is rejected.	Name boundary test	Met
AC-004	Invalid name characters are rejected.	Invalid character test	Met
AC-005	Invalid email format is rejected.	Email validation test	Met
AC-006	Phone number with fewer than 10 digits is rejected.	Phone boundary test	Met
AC-007	Phone number with more than 10 digits is rejected.	Phone boundary test	Met
AC-008	Non-numeric phone number is rejected.	Numeric validation test	Met
AC-009	Valid 10-digit phone number is accepted.	Valid phone test	Met
AC-010	Weak password is rejected.	Password security test	Met
AC-011	Strong password is accepted.	Strong password test	Met
AC-012	Valid student can be created.	Student Add test	Met
AC-013	Invalid student creation is blocked.	Negative Add test	Met
AC-014	Invalid student update is blocked.	Negative Edit test	Met
AC-015	Valid student update is accepted.	Student Edit test	Met
AC-016	Direct API invalid request is rejected.	Backend/API validation test	Met
AC-017	Existing Edit/Delete operations continue to work.	CRUD regression test	Met
AC-018	Existing authentication/authorization continues to work.	Access-control regression test	Met

5. Evidence

A maximum of five relevant evidence items can be attached to demonstrate the main CR-004 validation, CRUD, and security requirements.

Evidence 01 – Frontend Validation

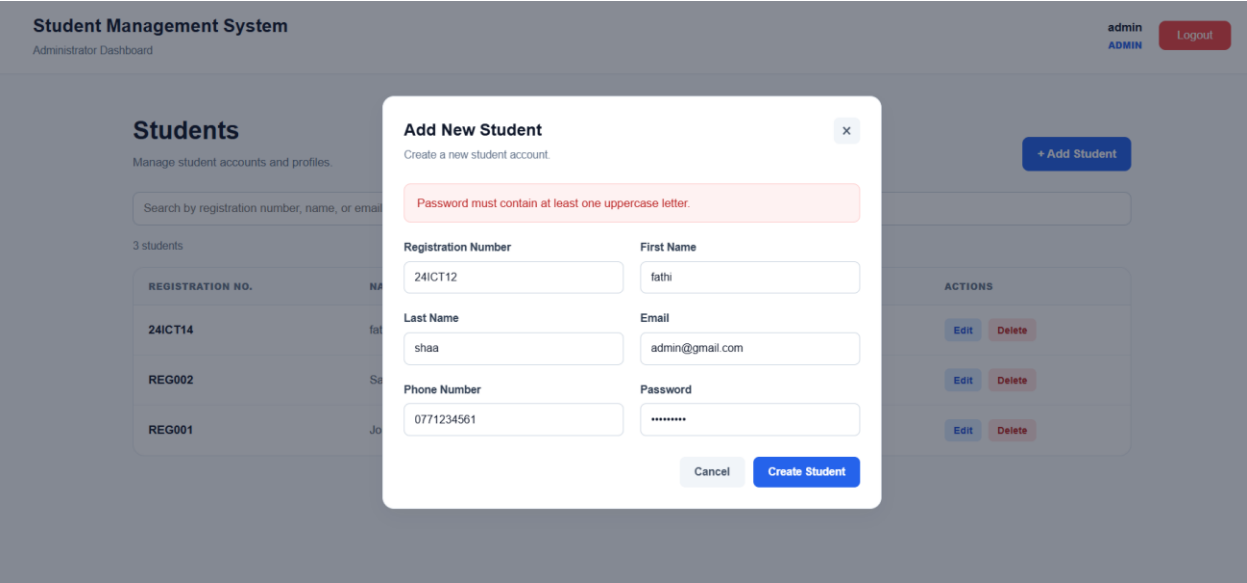
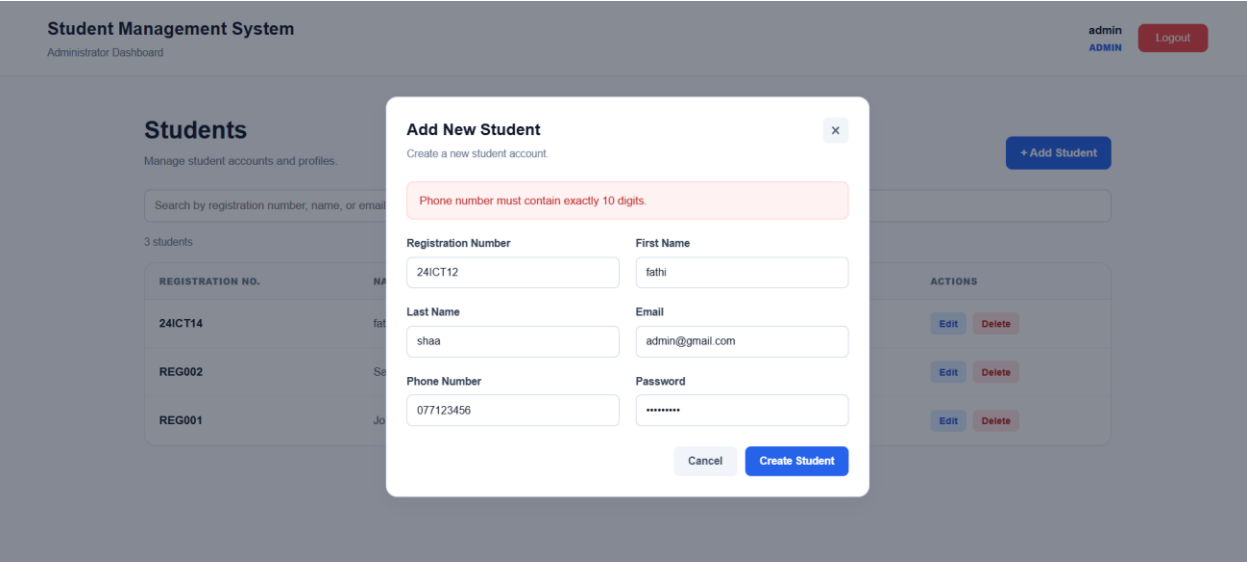
Related: AC-001, AC-002, AC-003, AC-004, AC-005, AC-006, AC-007, AC-008, AC-010

The screenshot displays the 'Add New Student' modal in the Student Management System. The modal contains a form with the following fields: Registration Number (24ICT12), First Name (Ab), Last Name (12), Email (admin@gmail.com), Phone Number (077123456), and Password (masked with asterisks). A red error message at the top of the form states: 'First name must contain at least 3 characters.' The background shows the 'Students' section of the dashboard with a table of existing students and a search bar.

REGISTRATION NO.	NAME
24ICT14	fath
REG002	Se
REG001	Jo

The screenshot displays the 'Add New Student' modal in the Student Management System. The modal contains a form with the following fields: Registration Number (24ICT12), First Name (fathi), Last Name (12), Email (admin@gmail.com), Phone Number (077123456), and Password (masked with asterisks). A red error message at the top of the form states: 'Last name must contain at least 3 characters.' The background shows the 'Students' section of the dashboard with a table of existing students and a search bar.

REGISTRATION NO.	NAME
24ICT14	fath
REG002	Se
REG001	Jo



Student Management System
Administrator Dashboard

admin
ADMIN Logout

Students

Manage student accounts and profiles.

Search by registration number, name, or email...

3 students

REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS
24ICT14	fath			Edit Delete
REG002	Sa			Edit Delete
REG001	Jo			Edit Delete

Add New Student
Create a new student account.

Registration Number: 24ICT12

First Name: shanthi

Last Name: kumar

Email: admin

Phone Number: 0787789087

Please include an '@' in the email address. 'admin' is missing an '@'.

Cancel Create Student

Figure 1: Student Add/Edit form showing validation errors for invalid field input.

Evidence 02 – Valid Student Creation

Related: AC-009, AC-011, AC-012

Student Management System
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admin
ADMIN Logout

Students

Manage student accounts and profiles.

Search by registration number, name, or email...

4 students

REGISTRATION NO.	NAME	EMAIL	PHONE	ACTIONS
24ICT12	shanthi kumar	test@gmail.com	0787789087	Edit Delete
24ICT14	fathi Shimla	admin@gmail.com	0789876431	Edit Delete
REG002	Sarah Fernando	sarah@example.com	0723456789	Edit Delete
REG001	John Perera	john@example.com	0712345678	Edit Delete

+ Add Student

Figure 2: Student Add form showing valid 10-digit phone number, strong password, and successful creation.

Evidence 03 – Edit Validation / Valid Update

Related: AC-014, AC-015, AC-017

The screenshot displays the 'Student Management System' interface. In the background, there's a sidebar with 'Students' and a search bar. The main area shows a list of students. Overlaid on this is the 'Edit Student' modal. The modal has a title 'Edit Student' and a subtitle 'Update student information.' Below this, a red error message box says 'Phone number must contain exactly 10 digits.' The form fields are: 'Registration Number' (24ICT12), 'First Name' (shanithi), 'Last Name' (kumar), 'Email' (test@gmail.com), and 'Phone Number' (078778908). At the bottom of the modal are 'Cancel' and 'Update Student' buttons.

Figure 3: Student Edit operation showing invalid update blocked and valid update completed.

6. Final Result

Item	Result
Requirements	Met
Implementation	Completed
Testing	Passed
Acceptance Criteria	All Met
Existing CRUD	Verified
Final Status	Accepted – Implementation Completed

6.1 Final Summary

CR-004 was implemented as a frontend and backend validation enhancement for the Student Add and Edit operations. The implementation validates required fields, registration number, first name, last name, email address, phone number, and new-student password. First and last names require at least three characters and valid characters, phone numbers require exactly 10 numeric digits, and new passwords require at least 8 characters including uppercase, lowercase, number, and special character. Client-side validation provides immediate feedback and server-side validation prevents invalid direct API requests. Existing Add, Edit, Delete, authentication, authorization, and password hashing functionality was preserved. The defined 18 test cases passed and all 18 acceptance criteria were met.

6.2 Issues / Limitations

No significant implementation issues identified.

Final submission should include the actual screenshots/API evidence and, if required by the project process, the actual commit/review details.

Field	Details
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Date	8 October 2026
Signature	

~ END OF REPORT